

A Lightweight End-to-End Three-domain Feature Fusion Network for Motor Imagery Decoding

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Abstract—To decode Motor Imagery EEG signals (MI-EEG), most studies have increasingly complicated network models and parameters without fully considering EEG characteristics, thereby limiting advancements in Brain-Computer Interface (BCI) systems and classification performance. To address these issues, we propose a lightweight end-to-end tri-domain feature fusion network named LTDFNet. Firstly, we introduce an Attention-based Spatio-temporal Convolution module (ABST) to extract low-dimensional spatio-temporal features from EEG. This module employs a lightweight Squeeze-and-Excitation (SE) attention mechanism to enhance the model's perception of crucial information. Secondly, Temporal Domain Convolutional (TDC) and Frequency Domain Convolution (SDC) modules utilize Temporal Convolutional Networks (TCN) and Fast Fourier Transform (FFT) to respectively learn high-dimensional temporal and frequency domain information. Finally, the Feature Fusion (FF) module integrates low-dimensional spatio-temporal features and high-dimensional temporal-frequency features effectively through learnable parameters. LTDFNet is trained with joint constraints of Softmax loss and Center loss functions to achieve optimal inter-class separation and intra-class compactness, thereby enhancing overall model performance. This study conducts extensive experimental validation on BCI Competition IV datasets 2a (BCI 2a) and 2b (BCI 2b). LTDFNet achieves classification accuracies of 76.89% (kappa score: 0.692) and 85.22% (kappa score: 0.704) on the BCI 2a and BCI 2b datasets, respectively. Compared to other high-performance decoding methods, LTDFNet utilizes only 20,576 parameters, balancing network scale and decoding performance requirements.

Keywords—Lightweight, Feature Fusion, Motor Imagery, Brain-Computer Interface

I. INTRODUCTION

Currently, EEG-based Brain-Computer Interface (BCI) paradigms include Steady-State Visually Evoked Potentials (SSVEP) [1], Event-Related Potentials (ERP) [2], and Motor Imagery (MI) [3]. Among these, MI-based BCIs have garnered significant attention due to their ability to activate specific motor regions of the brain without external stimulation. By decoding MI-EEG signals to discern the motor intentions of users and translating these into commands for controlling and manipulating external devices, it empowers individuals with motor impairments to operate assistive tools such as wheelchairs and prostheses [4-5], while simultaneously supporting motor rehabilitation training for stroke patients [6-7]. Accurate decoding of MI-EEG signals is

therefore crucial for these applications. However, EEG signals are influenced by various factors, including low signal-to-noise ratio (SNR), non-stationarity, and low spatial resolution, as well as significant individual differences, which complicate precise signal decoding [8-9].

Currently, deep learning (DL) has made significant strides in the MI-BCI field. Convolutional Neural Networks (CNNs) [10-12], among the most commonly used architectures, play a critical role in MI-BCI research. To further enhance the classification performance and robustness of MI-BCI systems, researchers are continuously expanding network sizes [13]. For example, the CP-MixedNet model proposed by Li et al. [14] extracts spatial and temporal representations from MI-EEG using channel projections and captures high-dimensional features across multiple time scales with mixed-scale convolutions, achieving 836K parameters. As research progresses, the increasing parameter count can limit the practical application of MI-BCI systems in real-world scenarios and hinder their development [15]. Additionally, the difficulty in collecting MI-EEG data results in generally small training datasets, making large network sizes prone to overfitting, which impedes improvements in model decoding and classification performance [16-17].

Considering the impact of model complexity and parameter count on MI-BCI, researchers have proposed lightweight DL networks for decoding EEG signals. Lawhern et al.[18] introduced a classic compact and lightweight network known as EEGNet. This network uses depthwise convolutional and separable convolutional layers to minimize the number of parameters. However, shallow networks may not fully extract feature information, leading to less accurate EEG signal decoding. Consequently, researchers have developed improved network models based on EEGNet to enhance classification performance and generalization capability [19]. For instance, Ingolfsson et al.[20] proposed EEG-TCNet, which incorporates temporal convolutional networks (TCNs) [21] to better leverage temporal information. Building upon EEG-TCNet, Musallam et al.[22] introduced an enhanced version called TCNet-Fusion. This model adds fusion layers to reduce feature information loss and creates richer feature maps to improve model performance. Bria et al. [23] proposed Sinc-EEGNet, which utilizes sinc convolutions [24] in its initial layer to extract frequency-domain information from EEG signals. sinc convolutions have fewer parameters compared to standard convolutions. Although these advancements have addressed some issues present in

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EEGNet, MI-EEG decoding remains constrained. This is because information related to movement in MI-EEG signals is often embedded in both the time and frequency domains, and the electrode positions during EEG signal recording can also impact the recognition of movement-related information [25].

Although lightweight networks like EEGNet reduce training time and enhance practical usability compared to larger-scale networks, they often focus on extracting feature information from either spatio-temporal, temporal-frequency, or spatio-frequency domains, neglecting the complementary relationships and dependencies among temporal-frequency-space domains[11][26][27]. Furthermore, the importance of feature fusion is frequently overlooked[28]. These issues hinder effective and comprehensive utilization of MI-EEG feature information, thereby challenging the enhancement of classification performance.

To address these limitations, this paper proposes a lightweight convolutional neural network based on the fusion of spatio-temporal-frequency domain features, named LTDFNet. Compared to existing methods, this model achieves high decoding performance with only 20,576 parameters while balancing parameter efficiency and performance. The main contributions of this paper are as follows:

- We propose a lightweight neural network model named LTDFNet for learning and fusing low-dimensional spatio-temporal information and high-dimensional temporal-frequency information in EEG signals to improve the performance of MI classification tasks.
- We introduce a sliding window in the convolution module to increase the amount of MI-EEG data and alleviate the overfitting problem caused by limited EEG samples.
- The LTDFNet model is trained using a combination of the Softmax function and center loss to obtain discriminative inter-class and compact intra-class features. The effectiveness of the proposed method was evaluated using two datasets. On the BCI 2a dataset, the classification accuracy was 76.89% (kappa score: 0.692). On the BCI 2b dataset, the

classification accuracy was 85.22% (kappa score: 0.704).

II. METHOD

The proposed LTDFNet model consists of four main modules: Attention-Based Spatio-temporal Convolution (ABST), Temporal Domain Convolution (TDC), Frequency Domain Convolution (SDC), and Feature Fusion (FF) modules. The specific structure is illustrated in Fig. 1. In the ABST module, low-dimensional spatio-temporal features are extracted using convolutional layers, enhanced by a Squeeze-and-Excitation (SE) [30] attention mechanism to improve sensitivity. This approach enables precise capture and focus on the most relevant features while reducing the impact of redundant information. To enhance model capacity and accuracy, spatio-temporal feature outputs are segmented into multiple windows, which are separately fed into the TDC and SDC modules for parallel extraction of high-dimensional temporal and frequency features. The TDC module employs Temporal Convolutional Networks (TCN) to extract long-term dependency features from time-series data, leveraging deeper information layers. The SDC module initially transforms time-series into frequency features using Fast Fourier Transform (FFT), followed by decoding through convolutional layers. In the FF module, outputs from the ABST module and parallel TDC and SDC modules are fed into fully connected layers and fused using learnable tensors. Finally, training combines the Softmax function with a center loss method to achieve inter-class separation and intra-class compactness of features.

A. Preprocessing and Input Representation

To fully extract the temporal, frequency, and spatial information inherent in the raw MI-EEG signals, this study refrains from conducting any preprocessing on the data. Instead, it directly inputs all frequency bands and channel information of the raw MI-EEG signals into the proposed LTDFNet model.

In this study, the MI-EEG signal $X_i \in R^{C \times T}$ is used as the input to LTDFNet, where C represents the number of channels (i.e., EEG electrodes), and T denotes the number of sampling points. LTDFNet aims to map the input signal X_i to its corresponding class y_i , where $y_i \in \{1, \dots, n\}$ denotes the class label of the input signal, and n represents the total

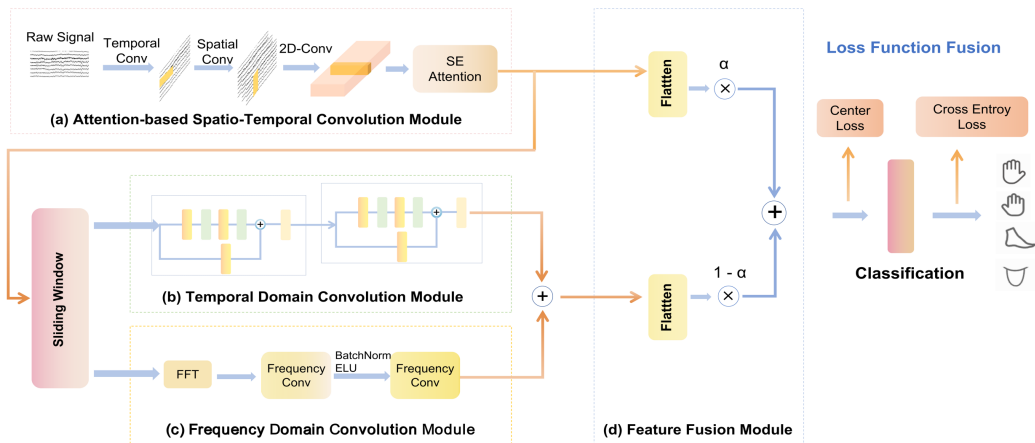


Fig. 1. Components of the proposed LTDFNet model

number of classes. This research evaluates the model performance using two datasets: for the BCI 2a dataset, $T = 1125$ time points, $C = 22$ electrode channels, and $n = 4$ classification categories; for the BCI 2b dataset, $T = 1000$ time points, $C = 3$ electrode channels, and $n = 2$ classification categories.

B. Attention-based Spatio-Temporal Convolution Module

In the original MI-EEG signals, there are numerous time, frequency, and spatial features that are difficult to manually define. LTDFNet first utilizes the ABST module to extract low-dimensional spatiotemporal features. Unlike EEGNet's architecture, this module employs 2D convolutions instead of separable convolutions and integrates the SE attention mechanism, enhancing the model's classification performance.

The ABST module consists primarily of three convolutional layers and an SE attention mechanism, as illustrated in Fig. 1(a). Initially, temporal convolution is applied to MI-EEG signals to extract temporal information using F_1 filters of size $(1, K_A)$, where K_A is set to one-fourth of the sampling rate, capturing temporal features above 4Hz. The second layer employs depthwise convolution on the temporal feature maps to extract spatial information, utilizing F_2 filters of size $(C, 1)$, where F_2 equals $F_1 \times D$. Here, D denotes the number of filters connected to each temporal feature map in the preceding layer, typically set to 2 based on previous experience. F_2 determines the output dimensionality of this module, varying with the specific number of EEG channels C across different datasets. Subsequently, average pooling of size $(1, 8)$ reduces the sampling frequency to approximately 32Hz. These layers collectively transform EEG signals into a combined representation of time and space. The third convolutional layer employs 2D convolution to further extract feature information, followed by average pooling of size $(1, P)$ to reduce the sampling frequency to approximately $32/P$ Hz. Batch Normalization (BN) is applied after each convolutional layer to accelerate model training and enhance generalization.

The attention mechanism mimics human brain behavior by selectively focusing attention on a few critical pieces of information while disregarding others. Integrating attention mechanisms with deep learning models helps automatically prioritize and learn important parts of vast datasets or features, thereby enhancing classification performance. Therefore, incorporating the SE attention mechanism into the ABST module aims to improve the model's sensitivity and expressive power towards spatiotemporal features, enabling it to focus on crucial information. Moreover, the SE attention module has fewer parameters, making it advantageous for achieving

lightweight networks. Its structure, illustrated in Fig. 2, primarily consists of the following three steps.

The first step involves the Squeeze operation, where the feature maps $U \in R^{H \times W \times C}$ extracted from the preceding layer are transformed using global average pooling to integrate global contextual information. The formula for this operation is as follows:

$$z_c = F_{sq}(U_c) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W u_c(i, j) \quad (1)$$

where $U_c \in R^{H \times W}$, $c \in [1, 2, \dots, C]$ represents the feature map of channel c in U . $U_c(i, j)$ denotes a data point within U_c . The Squeeze operation $F_{sq}(\cdot)$ compresses the two-dimensional feature U_c into the real number z_c .

Next, the Excitation operation is performed, which determines the weights for different channels using two fully connected layers. The formula for this operation is as follows:

$$s_c = \sigma(W_{F2}, \delta(W_{F1}, z)) \quad (2)$$

Where W_{F1} and W_{F2} denote the weight matrices of two fully connected layers, the first of which is reduced by a factor of r to decrease computational load. $\sigma(\cdot)$ represents the Sigmoid function, and $\delta(\cdot)$ denotes the ReLU function. Finally, the Scale operation is applied, defined by the following formula:

$$\tilde{x}_c = F_{scale}(u_c, s_c) = s_c u_c \quad (3)$$

$F_{scale}(\cdot)$ represents the process of weighting the channel-wise information outputted by Excitation onto the previous features through multiplication, thereby obtaining feature maps under different weights. Through these three steps, SE enhances the channel-wise features of the input feature map without altering its size.

After the ABST module, the model has learned and extracted shallow temporal-spatial feature information. To enhance the training data volume and classification accuracy, the time series is divided into m windows using a sliding window approach. However, to avoid significantly increasing computational burden, the sliding windows are integrated within the ABST module, enabling convolution operations to be computed just once over the m windows. The window length W_L is determined jointly by the length T_L of the time series and the number m of windows.

C. Temporal Domain Convolution Module

After obtaining temporal-spatial feature information from the ABST module, it becomes evident that these features are shallow and low-dimensional, insufficient for fully harnessing the wealth of information contained in EEG data. Therefore, to comprehensively extract and utilize deep, high-dimensional feature information, the TDC module further extracts profound temporal insights. This is achieved by introducing TCN to capture long-term dependencies in the time-series data.

TCN is a CNN-based model that differs from traditional CNNs in several key aspects: causal convolutions, dilated convolutions, and stacked residual blocks. Firstly, causal convolutions are designed to prevent information from future time steps influencing past predictions; hence, the output at time t depends only on inputs from time t and earlier. However, using only causal convolutions may necessitate deeper networks or larger convolutional kernels to increase receptive

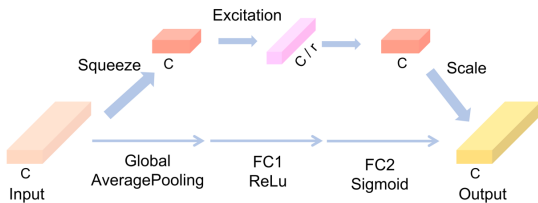


Fig. 2. The Squeeze-and-Excitation (SE) Block

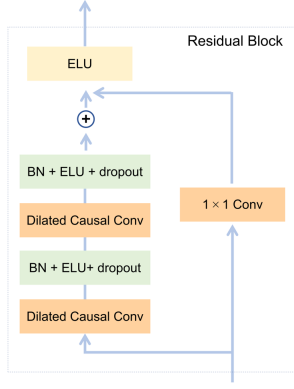


Fig. 3. The TCN Residual Block

fields. To address this, TCN introduces dilated convolutions, which expand the receptive field exponentially by using an increasing dilation factor d . TCN is composed of stacked residual blocks, where each block consists of two layers of dilated causal convolutions, BN, ELU, and a dropout layer, as depicted in Fig. 3. Skip connections add the input to the output feature maps and ensure that the depths of input and output are identical; if not, a 1×1 convolution is used for linear transformation.

By stacking residual blocks, the receptive field size in TCN also exponentially increases with each subsequent block due to the exponentially growing dilation factor d . The receptive field size (RFS) of TCN is determined by the following factors:

$$RFS = 1 + 2 \cdot (K_T - 1) \cdot (2^L - 1) \quad (4)$$

where K_T denotes the convolution kernel size in TCN residual blocks, and L represents the number of stacked residual blocks.

In LTDFNet, TCN residual blocks differ slightly from standard TCN and share the same structure as the residual blocks in EEG-TCNet, using Batch Normalization (BN) instead of Weight Normalization (WN) after convolution for improved accuracy. Setting K_T to 4 and L to 2 results in an RFS of 19. Therefore, the length of the time series input to the TDC module from the ABST module should not exceed 19 to avoid losing remaining time information.

D. Frequency Domain Convolution Module

To convert time-series data into frequency-domain signals, we introduced the FFT algorithm within the SDC module, applying it to the output of the ABST module to better decode frequency components in MI-EEG. Traditional Fourier Transform (FT) algorithms are computationally intensive, limiting efficiency gains. FT's complexity primarily depends on signal length and struggles to achieve computational efficiency. To overcome these limitations, researchers developed the FFT algorithm, which significantly enhances computational efficiency through techniques like divide-and-conquer strategies and butterfly operations. Therefore, we chose to employ FFT within the SDC module for signal frequency-domain transformation, enabling more effective analysis of frequency characteristics in MI-EEG.

$$U' = FFT(U) \quad (5)$$

To obtain frequency-domain signals using the FFT algorithm, we employ convolutional layers to extract frequency information. To accelerate convergence and enhance the model's generalization performance, we integrate Batch Normalization (BN) layers immediately after the convolutional layers.

The output of the ABST module undergoes feature extraction and learning in the TDC and SDC modules, acquiring deep, high-dimensional temporal and frequency characteristics. This further enhances the model's generalization performance and classification accuracy. By summing the output features from the TDC and SDC modules, we integrate information from different feature spaces, using it as the overall input to the fusion module. This enhances the model's expressive power and classification performance.

E. Feature Fusion Module

The MI-EEG signals are initially processed by the ABST module to extract low-dimensional spatio-temporal features. Subsequently, these low-dimensional features are separately inputted into the TDC and SDC modules to extract high-dimensional temporal and frequency features. The low-dimensional spatio-temporal features and high-dimensional temporal-frequency features are then simultaneously fed into the feature fusion module.

Specifically, the FF module receives the output features from the ABST module and the fused output features (T-SFF) from the TDC and SDC modules as its two inputs. By fully utilizing multiple different input features, the model not only reduces feature loss but also enhances its capability to integrate feature information. The output information from the ABST module and T-SFF undergo processing through fully connected layers to yield E_1 and E_2 , respectively. Subsequently, the Sigmoid activation function maps E_1 and E_2 to the (0,1) range, ensuring the effectiveness of the fusion coefficients. Finally, E_1 and E_2 are weightedly fused using learnable parameters α to obtain the result of the FF module.

$$E = f(\alpha) \cdot E_1 + (1 - f(\alpha)) \cdot E_2 \quad (6)$$

where $f(\cdot)$ represents the Sigmoid activation function.

F. Loss Function Fusion

LTDFNet employs two loss functions to jointly constrain model training for improved overall performance. By combining Softmax loss and center loss functions, it achieves a balance between inter-class separation and intra-class cohesion.

In the final stage of the LTDFNet model, Softmax is employed as the classifier to transform the output into a probability distribution, as illustrated below:

$$\tilde{y}_n = \text{Softmax}(u_i) \quad (7)$$

where u_i represents the output features. The cross-entropy is then computed to measure the distance between the actual output distribution and the true distribution. Training utilizes the Adam optimizer to minimize cross-entropy, thereby maximizing inter-class separation. The process of computing cross-entropy is as follows:

$$L_{CE} = - \sum_{k=1}^m y_k \log(\tilde{y}_k) \quad (8)$$

where y_k refers to the label value of the true sample, while $\tilde{y}_k$ denotes the actual output, i.e., the probability values obtained after Softmax computation.

This study utilizes the Softmax loss function to achieve differentiation between classes, complemented by the center loss function to ensure feature distinctiveness and discriminability. The computation formula for the center loss function is as follows:

$$L_C = \frac{1}{2N} \sum_{i=1}^N \|u_i - v\|_2^2 \quad (9)$$

where N represents the batch size of the mini-batch, u_i denotes the output features, and v represents the corresponding center of the i -th category. Empirically, the learning rate for the center loss function is typically set to 0.5.

Thus, the final form of the loss function is as follows:

$$L_{total} = L_{CE} + \lambda \cdot L_C \quad (10)$$

To maximize feature discrimination, LTDFNet is optimized jointly using two loss functions. Here, λ represents a weighted hyperparameter used to balance the importance of the two loss functions.

III. EXPERIMENTAL RESULTS

A. Datasets

In this study, we validate the classification performance of the proposed LTDFNet model using two well-established BCI competition datasets. The detailed information of these two datasets is as follows:

The first dataset used is the BCI Competition IV Dataset 2a (BCI 2a)[31], which recorded MI-EEG signals from 9 subjects using 22 electrodes sampled at 250 Hz. Each subject participated in two sessions over two days, with each session comprising 288 EEG trials covering four MI tasks (left hand, right hand, tongue, and feet). Training data for the model were obtained from all trials of the first session, while testing data were sourced from the second session. All data used for the model were collected within 4 seconds after the onset of the cue stimulus. For further details regarding this dataset, please visit <https://www.bbc.de/competition/iv/#dataset2a>.

The second dataset utilized is the BCI Competition IV Dataset 2b (BCI 2b)[32], which recorded MI-EEG signals from 9 subjects using 3 electrodes (C3, Cz, and C4) sampled at 250 Hz. Each subject underwent experiments across five sessions, totaling 720 trials from two MI tasks (left hand, right hand). The first two sessions included 120 trials each without feedback, while the last three sessions included 160 trials each with

feedback displayed as a smiling face on the screen. Training data for the model were derived from all trials of the first three sessions, while testing data were sourced from the last two sessions. All data used for the model were collected within 4 seconds after the onset of the cue stimulus. For further details regarding this dataset, please visit <https://www.bbc.de/competition/iv/#datasets>.

B. Experimental Details

In this study, we trained and tested our model using the TensorFlow 2.0 framework and Python 3.7. We employed Adam as the primary optimizer to fine-tune model parameters, setting the learning rate to 0.001 and a batch size of 64. Additionally, during training, we implemented the ReduceLROnPlateau strategy to dynamically reduce the learning rate and enhance model performance.

When evaluating the model performance, we utilized classification accuracy and Kappa score (k-score) as the primary metrics.

$$ACC = \frac{\sum_{i=1}^n TP_i / l_i}{n} \quad (11)$$

where TP_i represents the number of correctly predicted samples in class i , l_i denotes the total number of samples in class i , n represents the total number of classes. In datasets BCI 2b and BCI 2a, n is 2 and 4 respectively.

$$Kappa = \frac{1}{n} \sum_{o=1}^n \frac{P_o - P_e}{1 - P_e} \quad (12)$$

where P_o denotes the actual agreement rate, P_e represents the theoretical agreement rate, and n indicates the number of categories.

In this study, subject-dependent experiments were conducted on both datasets. The model was trained and tested in the following manner: training on the first session of BCI 2a and testing on the second session; training on the first three sessions of BCI 2b and testing on the last two sessions.

C. Classification Performance

We conducted extensive subject-dependent experiments, as shown in Table I, comparing networks of different scales on BCI 2a, including parameters, multi-domain information fusion, and classification accuracy. Tables II, III, and IV present detailed decoding accuracies, standard deviations, k-score, and parameter counts of these models on BCI 2a and BCI 2b.

TABLE I. Comparison of Different network sizes on BCI 2a Dataset

Method	#params	Multi-domain Information			Average
		time domain	spatial domain	frequency domain	
ConvNet[10]	47,364	√	√	-	65.97
EEGNet[18]	2,548	√	√	-	60.60
EEGTCNet[20]	4,096	√	√	-	65.70
TCNet_Fusion[22]	17,248	√	√	-	70.73
TS-SEFFNet[29]	282,000	√	√	√	74.71
CP-MixedNet[14]	836,000	√	√	-	67.17
LiteEEG-inception[33]	417,990	√	√	-	74.51
LiteEEG-SE-inception[33]	858,520	√	√	-	76.52
EEG Conformer[34]	789,816	√	√	-	78.66
LTDFNet(Ours)	20,576	√	√	√	76.89

TABLE II. Comparative Analysis of Classification Performance between LTDFNet and Lightweight Networks on BCI 2a Dataset

Subject	ConvNet[10]	EEGNet[18]	EEGTCNet[20]	TCNet_Fusion[22]	TSFF-Net[37]	LTDFNet (Ours)
A01	72.92	68.40	77.78	75.69	74.30	82.64
A02	46.53	50.40	47.92	57.64	56.30	69.44
A03	83.68	77.80	79.86	85.42	78.80	91.32
A04	52.08	47.20	57.64	59.38	51.70	74.31
A05	60.42	43.40	60.76	65.97	52.10	67.01
A06	42.01	45.80	49.31	56.60	50.30	63.89
A07	78.12	86.80	76.04	85.42	86.10	89.24
A08	79.86	55.90	74.31	72.92	61.50	79.86
A09	78.12	69.80	67.71	77.43	75.70	74.31
Average	65.97	60.60	65.70	70.72	65.20	76.89
Standard	15.00	15.60	11.57	10.69	13.60	9.06
k-score	0.546	-	0.543	0.610	-	0.692
#params	47,364	2,548	4,096	17,248	-	20,576

TABLE III. Comparative Analysis of Classification Performance between LTDFNet and Large-Scale Networks on BCI 2a Dataset

Subject	TS-SEFFNet[29]	CP-MixedNet[14]	LiteEEG-inception[33]	LiteEEG-SE-inception[33]	MBSTCNN-ECA-LightGBM[35]	EEG-Conformer[34]	LTDFNet (Ours)
A01	82.29	74.65	77.58	79.31	82.00	88.19	82.64
A02	49.79	53.47	64.65	57.75	61.00	61.46	69.44
A03	87.57	73.26	92.24	87.93	89.00	93.40	91.32
A04	71.74	70.14	60.34	62.93	63.00	78.13	74.31
A05	70.83	67.36	65.51	75.86	71.00	52.08	67.01
A06	63.75	48.96	59.48	59.48	64.00	65.28	63.89
A07	82.92	74.31	87.93	91.37	72.00	92.36	89.24
A08	81.53	72.92	86.20	91.37	79.00	88.19	79.86
A09	81.94	69.44	76.72	83.62	84.00	88.89	74.31
Average	74.71	67.17	74.51	76.52	74.00	78.66	76.89
Standard	11.32	8.88	11.79	12.62	9.53	14.43	9.06
k-score	0.663	0.562	-	-	0.650	0.7155	0.692
#params	282,000	836,000	417,990	858,520	-	789,816	20,576

Analysis of Table I reveals that most lightweight and large-scale neural networks only consider feature information from two domains, neglecting the complex characteristics of EEG signals. Furthermore, lightweight networks often exhibit lower decoding accuracy and rarely surpass the performance levels of large-scale networks. However, LTDFNet integrates temporal, spatial, and frequency domains, addressing the feature fusion shortcomings of existing networks of varying scales. Notably, LTDFNet achieves a remarkable accuracy of 76.89% using only the parameter 20,576, outperforming many large networks.

The experimental results on the BCI 2a dataset are presented in Table II. LTDFNet achieves consistently high average accuracy with a small standard deviation, outperforming several lightweight models by achieving peak accuracies across multiple subjects. The mean accuracy of LTDFNet is superior to that of ConvNet and EEGNet by 10.92% and 16.29%, respectively. Compared to EEGTCNet and the enhanced TCNet-Fusion based on EEGNet, our model

improves by 11.19% and 6.17% respectively. Compared to TSFF-Net incorporating CWT, our model outperforms it by 11.69%, with higher classification accuracy across eight subjects, demonstrating superior individual adaptability.

As shown in Table III, LTDFNet exhibits significant advantages over larger-scale network models on the BCI 2a dataset. Compared to TS-SEFFNet, our model improves accuracy by 2.18%. Moreover, LTDFNet requires 92.70% fewer parameters than TS-SEFFNet (282,000), with a smaller standard deviation indicating more robust accuracy across all participants. Our model achieves superior average accuracy and k-score values compared to CP-MixedNet by 9.72% and 23.13%, respectively, despite CP-MixedNet's higher parameter count of 836,000, which exceeds LTDFNet's by 97.54%. Compared to Inception-based LiteEEG-inception and LiteEEG-SE-inception, our model improves accuracy by 2.38% and 0.37%, with reduced standard deviations of 23.16% and 28.21%, respectively. Compared to LiteEEG-SE-inception, our model improves modestly, but the number of

TABLE IV. Comparative Analysis of Classification Performance between LTDFNet and Other Networks on BCI 2b Dataset

Subject	ConvNet[10]	EEGNet[18]	EEGTCNet[20]	TCNet_Fusion[22]	SHNN[36]	EEG-Conformer[34]	LTDFNet (Ours)
B01	70.94	77.50	71.56	71.25	83.33	82.50	76.25
B02	63.21	61.10	65.71	66.07	61.76	65.71	70.71
B03	72.19	63.10	81.25	80.94	58.33	63.75	81.56
B04	97.50	98.40	96.88	95.00	97.30	98.44	95.94
B05	84.69	96.60	84.38	95.31	91.89	86.56	93.75
B06	83.75	83.80	70.94	82.81	88.89	90.31	81.56
B07	85.31	84.40	88.75	90.62	86.11	87.81	93.12
B08	89.06	92.80	91.56	93.12	92.11	94.38	91.56
B09	81.56	88.40	80.00	86.56	91.67	92.19	82.50
Average	80.91	82.90	81.23	84.63	83.49	84.63	85.22
Standard	9.85	12.72	9.77	9.86	13.10	11.49	8.26
k-score	0.618	-	0.625	0.693	0.669	0.6926	0.704
#params	47,364	2,548	4,096	17,248	-	789,816	20,576

parameters is only 20,576, in contrast to LiteEEG-SE-inception's parameters of 858,520, which hampers training speed and practical usability. In comparison to MBSTCNNECA-LightGBM, the decoding accuracy of our model demonstrated an improvement of 2.89%, with seven subjects exhibiting superior classification accuracy in relation to that of the model. The LTDFNet model demonstrates a reduced standard deviation in comparison to the EEG Conformer model, with a 97.39% reduction in the number of parameters.

The experimental results on the BCI 2b dataset are shown in Table IV. LTDFNet achieved the best average classification accuracy and k-score with minimal standard deviation. Compared to classic networks ConvNet and EEGNet, our model demonstrated performance improvements of 4.31% and 2.32%, respectively. LTDFNet's decoding accuracy and k-score surpassed EEGTCNet and TCNet-Fusion. Compared to SHNN, our model is more concise and lightweight, with superior classification performance by 1.73% and a reduced standard deviation of 36.95%, indicating higher stability. In comparison to the EEG Conformer model, our proposed model demonstrates a higher average accuracy of 0.59%, accompanied by a notable reduction in both standard deviation and the number of covariates.

D. Ablation Study

In this section, we conducted ablation experiments to investigate the efficacy of each module within the LTDFNet model, and the impact of removing specific modules on MI classification performance on the BCI 2a dataset is presented in Table V. The results demonstrate that the TDC module and the sliding window have the most significant impact on the decoding performance. The removal of the TDC and sliding window modules resulted in a 14% and 7.25% reduction in average accuracy, respectively. In addition to the aforementioned two modules, each module also makes a contribution, including the removal of the SE attention mechanism, SDC and FF module from the LTDFNet model. The results of the study show that the corresponding average accuracies are reduced by 3.9%, 3.47% and 1.93%, respectively. Finally, when only the Softmax loss function was used, the classification accuracy of the model decreased by 1.97%. This indicates that the joint use of the two loss functions more effectively constrains the training of the model and enhances the capacity for feature classification.

IV. DISCUSSION

In this study, we propose an innovative lightweight and efficient LTDFNet model with a parameter count of 20,576, which significantly reduces the model size (Table I). The LTDFNet model has been designed with full consideration of the balance between network size and classification performance, which makes it more advantageous in accelerating the model inference process, improving training

efficiency, as well as enhancing real-world scenario adaptability. In comparison with alternative lightweight models, LTDFNet attains the highest classification accuracy and k-score value, while demonstrating the lowest standard deviation. This effectively addresses the limitations of lightweight models in feature information extraction. In comparison with the majority of large-scale models, LTDFNet not only demonstrates superior decoding performance but also exhibits a lower standard deviation.

Furthermore, LTDFNet fully exploits and makes use of the dependencies and complementarities among temporal, spatial and spectral information in EEG signals, which significantly improves the decoding performance. Firstly, in comparison to models that solely extract spatio-temporal, spatio-frequency or spatio-frequency domains, LTDFNet effectively addresses the limitations of two-domain models that are unable to fully utilise EEG signals by fusing three-domain feature information. Moreover, LTDFNet exhibits enhanced performance compared to models that extract information in the spatio-temporal-frequency domain. This is due to its distinctive approach, which integrates low-dimensional spatio-temporal features with high-dimensional temporal-frequency features from multiple perspectives.

Despite the aforementioned advances, there remain a number of limitations that require further investigation. Firstly, in comparison with the classical lightweight model, it is necessary to investigate further methods of reducing the network size without compromising the classification performance. Furthermore, given the relatively limited quantity of MI-EEG data, future research will aim to identify additional data enhancement strategies to address this challenge.

V. CONCLUSION

This paper proposes a high-performance and lightweight model, LTDFNet, for decoding MI-EEG signals. LTDFNet initially learns low-dimensional spatio-temporal representations, focusing on local spatio-temporal features. Subsequently, an SE attention mechanism is introduced to calibrate the feature map and enhance the network's sensitivity to critical information. Subsequently, high-dimensional temporal-frequency information is extracted with the objective of further exploiting the feature information present in the EEG signal. In conclusion, the FF module effectively integrates the low-dimensional spatio-temporal information and high-dimensional temporal-frequency information through learnable parameters. The LTDFNet model exhibits superior performance on the BCI 2a and BCI 2b datasets, as evidenced by its higher average accuracy, k-score, and parameter efficiency. In comparison with other high-performance models, the effectiveness and sophistication of LTDFNet are further verified.

TABLE V. Results of LTDFNet Ablation Experiments on BCI 2a Dataset

Removed module	Average	Standard	k-score
None	76.89	9.06	0.692
SEBlock	72.99	11.61	0.640
Sliding Window	69.64	11.29	0.595
TDC	62.89	13.40	0.505
SDC	73.42	12.49	0.646
FF	74.96	10.74	0.666
Loss Fusion	74.92	10.91	0.666

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